Applications of Information and Communication Technologies

Lecture-3

Today's Agenda

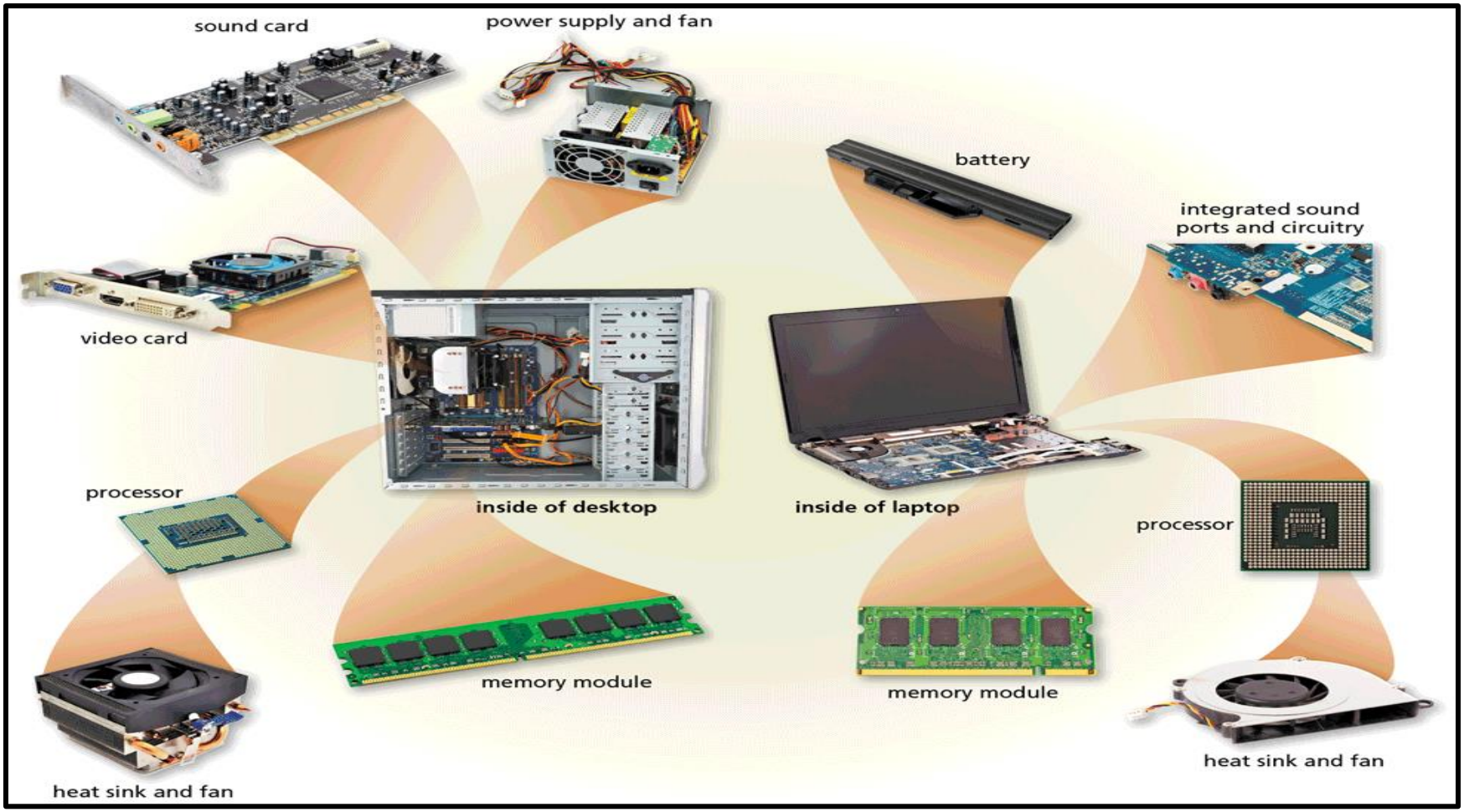
- Basic Operations and Components of a Generic Computer System
- Input, Processing, Output, Storage, Hardware, Software, Data, Users

Essential Computer Hardware

- A computer's hardware devices fall into following four categories

1. Processor
2. Memory
3. Input and output
4. Storage

Inside the Case

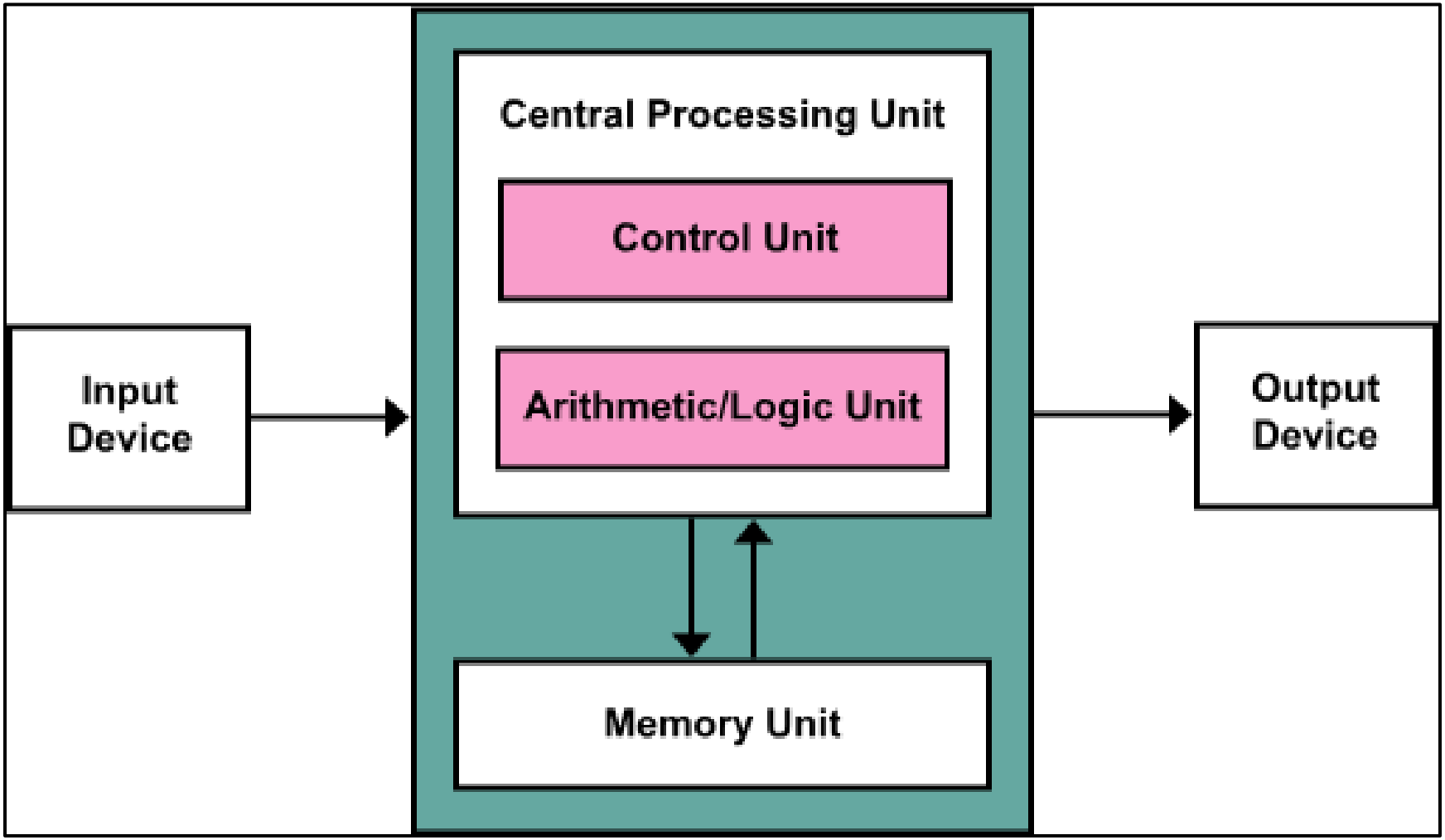


Von Neumann

- Mathematician,
- Physicist,
- Computer scientist
- Engineer
- Father of game theory



Von Neumann Architecture

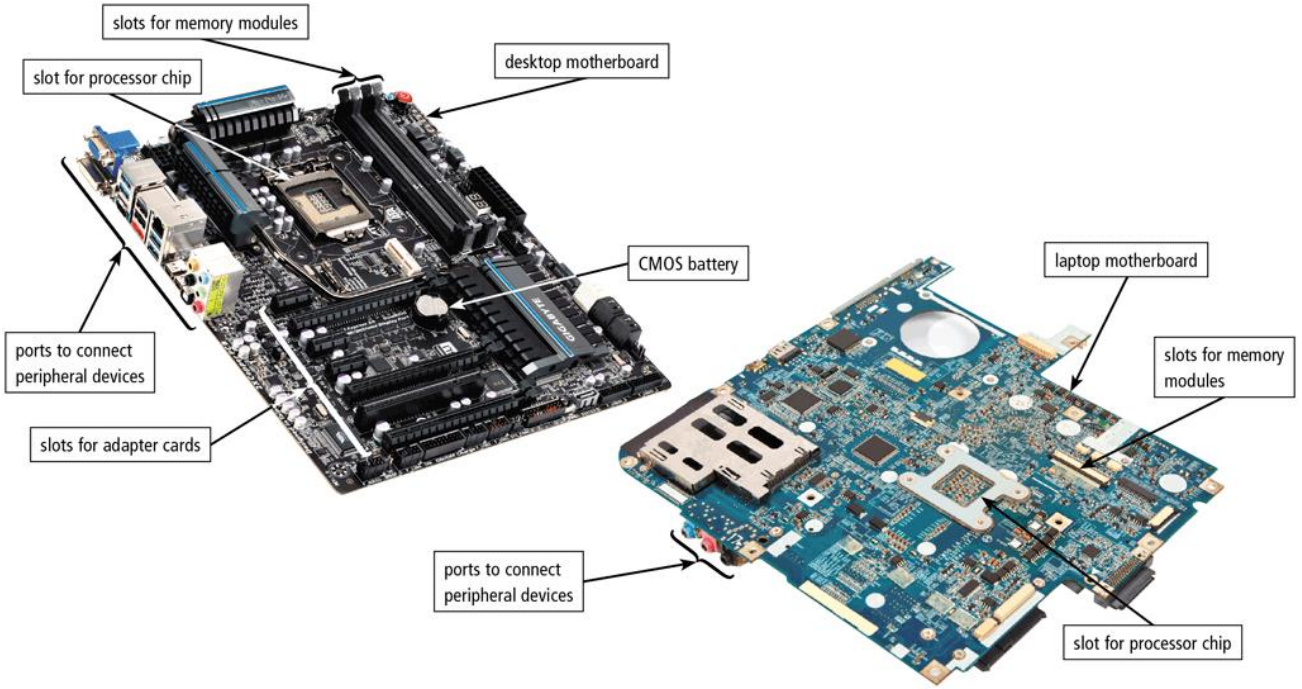


Von Neumann Architecture

- Input Unit stores information, encodes it and sends to the computer. Computer processes this information and gives output.
- ALU performs arithmetic and logical operations
- Control Unit controls Input, Output, AL and Memory units by sending Control Signals. It also controls data transfer between processor and memory.
- Memory Unit stores all the information in a group of memory cells called Binary Digits.

Inside the Case

- The **motherboard** is the main circuit board of the computer.



Motherboard

- Most important circuit board in system unit
- A communication medium for the entire computer system
- All components or devices of the computer are connected to motherboard
- Contain different chips
- A chip is a small semi conductor that contains integrated circuits on it
- An integrated circuit is a microscopic path that carries electrical current

Processors

- The **processor**, also called the **central processing unit** (**CPU**), interprets and carries out the basic instructions that operate a computer
- A **multi-core processor** is a single chip with two or more separate processor cores

– Processors contain a control unit and an arithmetic logic unit (ALU)

Processor

- Brain of the computer
- Performs all the operations on the data according to program's instruction
- It executes the program instructions and tells other parts of the computer what to do
- CPU **fetches** instructions of program from main memory and executes them one by one
- CPU also has the ability to decide which instruction will be executed next

Processor

- Speed of the CPU is measured in Mega Hertz or Giga Hertz
- CPU is divided into two parts or sub-units
  - Control Unit (CU)
  - Arithmetic Logic Unit (ALU)

Single core vs multi core

CONSIDER THIS

Are multi-core processors better than single-core processors?

Each processor core on a multi-core processor generally runs at a slower speed than a single-core processor, but multi-core processors typically increase overall performance. For example, although a dual-core processor does not double the processing speed of a single-core processor, it can approach those speeds. The performance increase is especially noticeable when users are running multiple programs simultaneously, such as antivirus software, spyware remover, email program, Internet messaging, media player, and photo editing software. Multi-core processors also are more energy efficient than separate multiple processors, requiring lower levels of power consumption and emitting less heat inside the case.

Processors

- The **control unit** is the component of the processor that directs and coordinates most of the operations in the computer
- The **arithmetic logic unit** (ALU) performs arithmetic, comparison, and other operations

What Is Input?

- Input is any data and instructions entered into the memory of a computer



What Is Output?

- Output is data that has been processed into a useful form



Memory

- **Memory** consists of electronic components that store three basic categories of items:

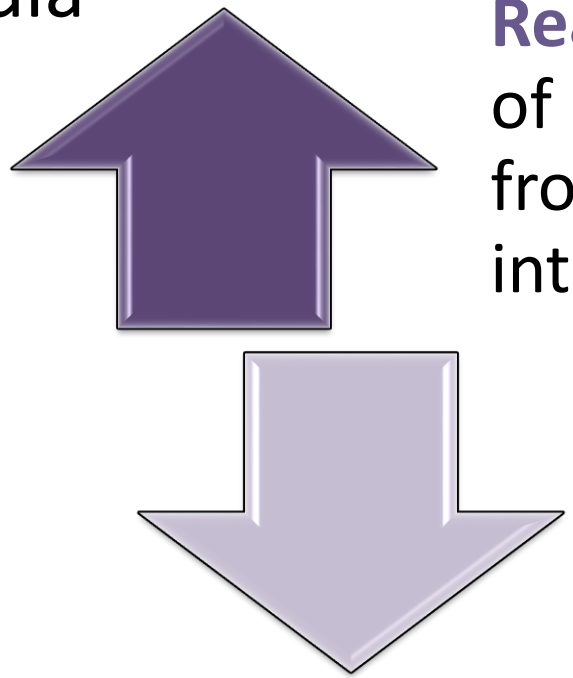
The operating system and other programs

Applications

Data being processed and the resulting information

Storage

- A **storage device** is the hardware that records and/or retrieves items to and from storage media



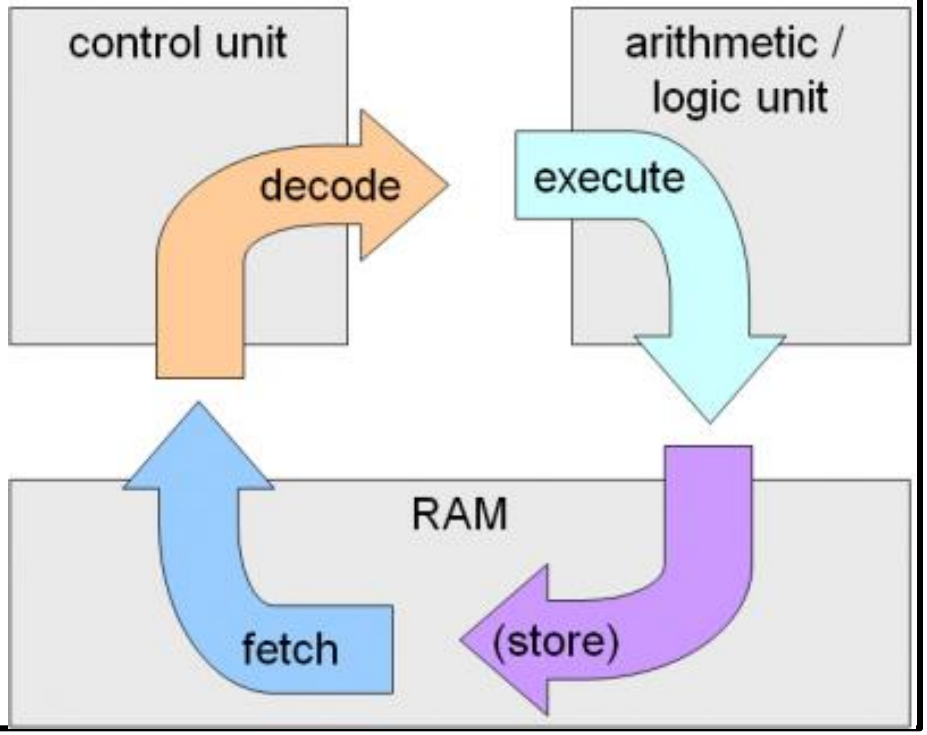
Reading is the process of transferring items from a storage medium into memory

Writing is the process of transferring items from memory to a storage medium

Machine Cycle

- For every instruction, a processor repeats a set of four basic operations, which comprise a machine cycle

1. Fetch
2. Decode
3. Execute
4. Store



System Clock

- Controls all functions of the computer using clock ticks
- The ticks of system are known as clock cycle
- Set the speed of CPU
- Computer clock can tick from millions to billions times in one second

System Clock

- The power of CPU is determined by the speed with which it processes data
- A CPU that has higher clock speed can process more instructions per second than a CPU with lower clock speed
- The speed at which the CPU executes instructions is called clock speed or clock rate
- Processor speed is measured in MHz and GHz
- Hertz means "machine cycle per second"

System Clock

CONSIDER THIS

Does the system clock also keep track of the current date and time?

No, a separate battery-backed chip, called the *real-time clock*, keeps track of the date and time in a computer. The battery continues to run the real-time clock even when the computer is off.

Central processing unit (CPU)



motherboard

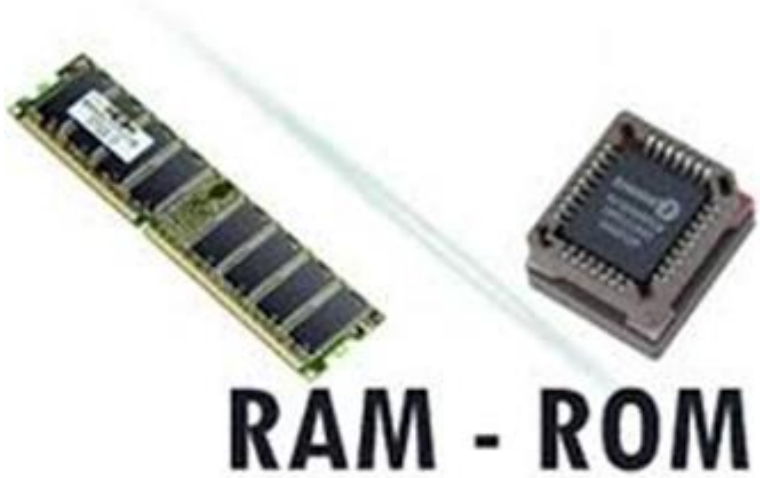


processor

- Refers to a computer’s processor
- Example: A "Pentium 4” system, for example, uses a Pentium 4 microprocessor as its CPU.

Memory Devices

- One or more sets of chips that store data and program instructions, either temporarily or permanently.
- Two most important memory devices are;
  - RAM
  - ROM



Random access memory (RAM)

- Volatile - Data is lost if the computer crashes or is turned off.
- Data is temporary - it is only there as long as the computer is on and it can be changed
- Data can be written to or read from.
- Also called main memory, or primary memory

Software

- Set of instructions that makes the hardware perform a particular set of tasks in particular order.

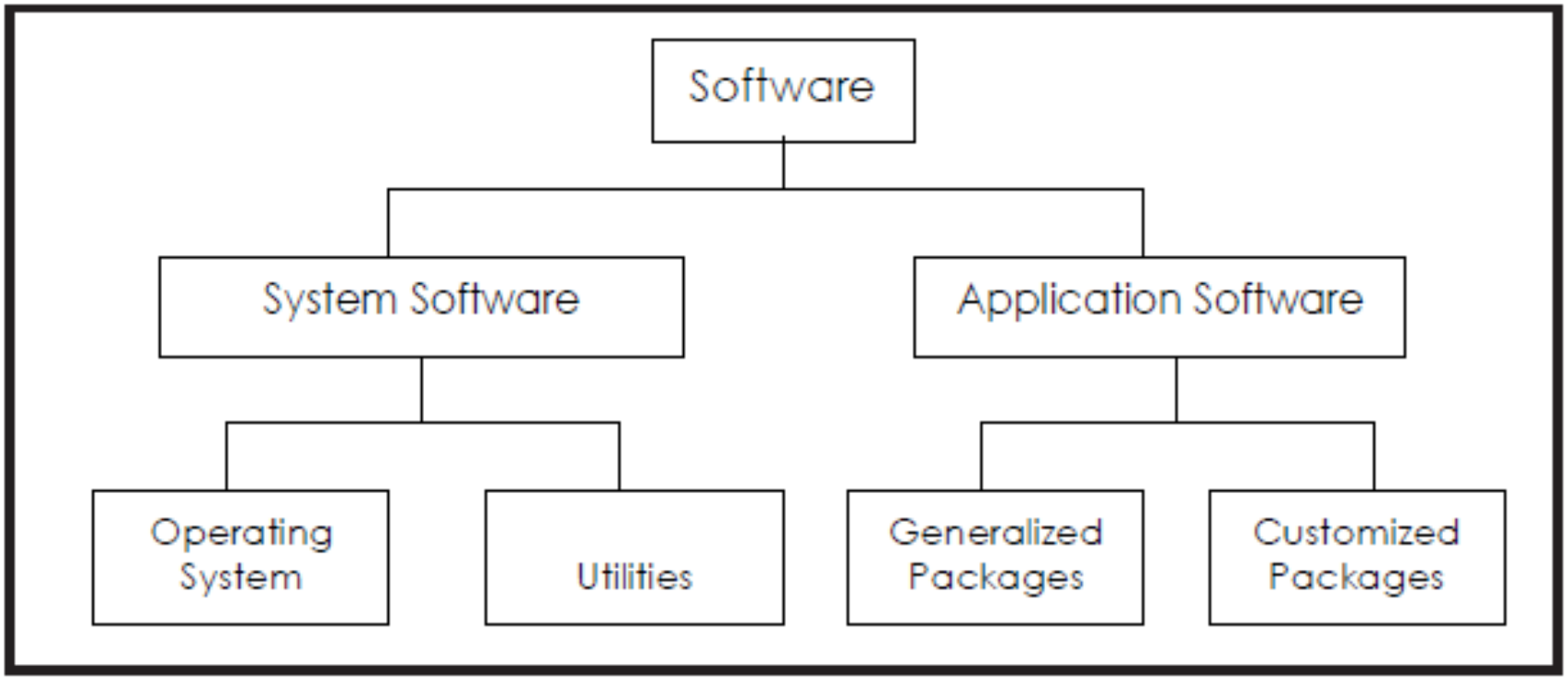
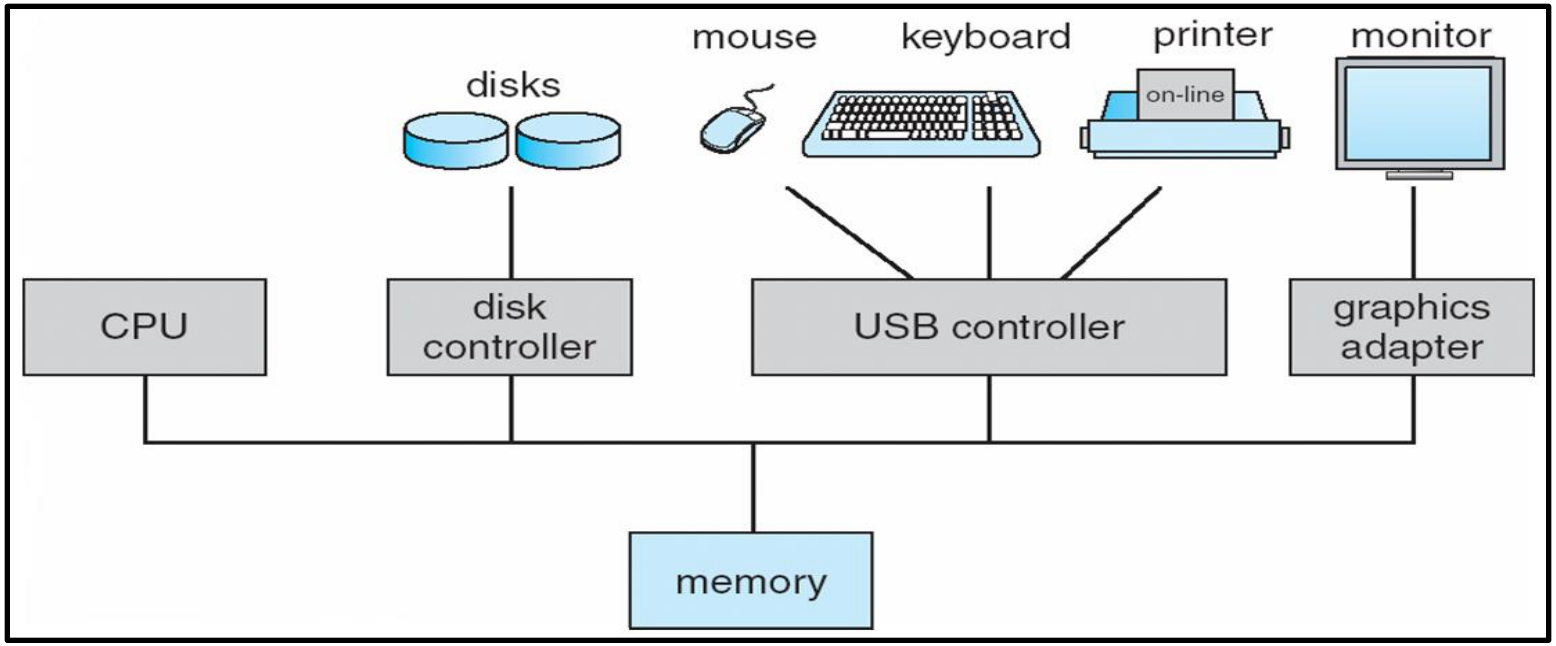


Image: Classification of Software

Computer-system operation

- One or more CPUs, device controllers connect through common bus providing access to shared memory
- Concurrent execution of CPUs and devices



Reference/Self Study

- Discovering Computers
  - Chapter-6